

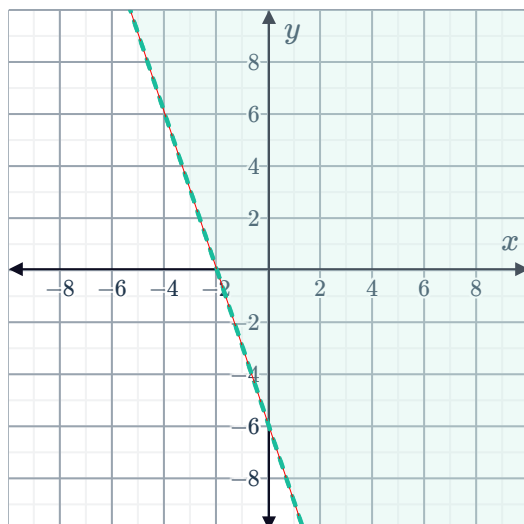
Linear inequalities



- 1
- | | | | |
|-----------------|----------------|----------------|-----------------|
| a A dashed line | b A solid line | c A solid line | d A dashed line |
| e A dashed line | f A solid line | g A solid line | h A dashed line |
- 2
- | | | | |
|--------------|-------------------|--------------------|----------------|
| a $x \geq 2$ | b $y \leq 5$ | c $y \geq -2x - 6$ | d $y < 2x + 6$ |
| e $y > -3x$ | f $y \leq 2x + 5$ | | |
- 3 No
- 4 Yes
- 5
- | | | |
|-----|-----|-----|
| a B | b C | c D |
|-----|-----|-----|

Graphing linear inequalities

- 6
1. Replace the inequality symbol with an equals sign and graph the linear equation.
 2. Choose a test point from one of the half planes that is not on the line. Substitute its coordinates into the inequality.
 3. If a true statement results, shade the half plane containing the test point, if false then shade the other half plane.
- 7
- | | |
|--------------------|------|
| a $x = -2, y = -6$ | b C |
| c | d No |



8

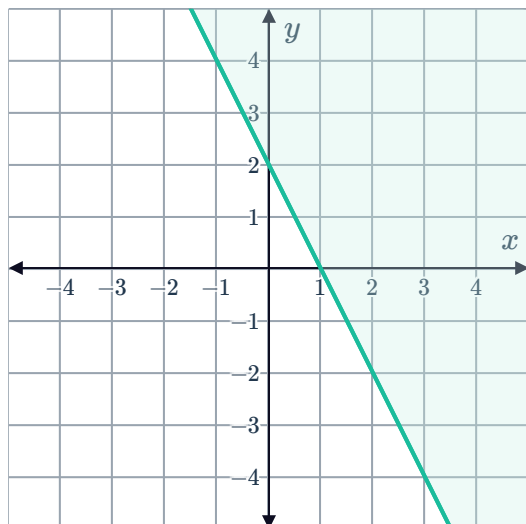
a $x = 1, y = 2$



b B

c

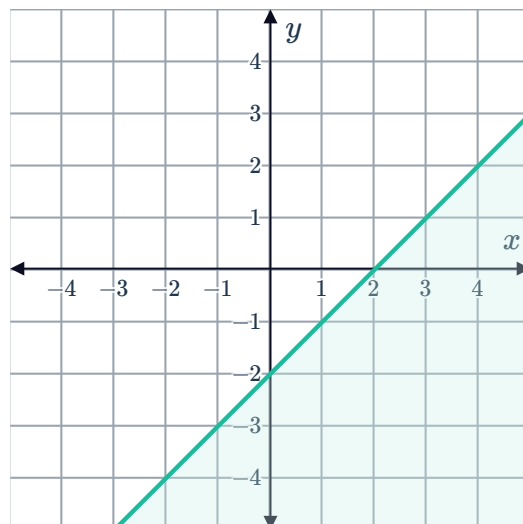
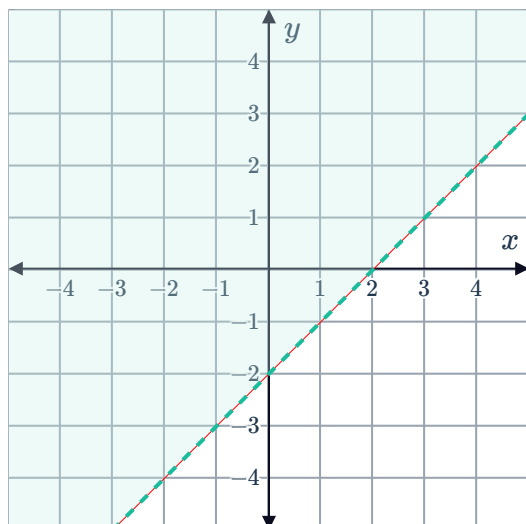
d Yes



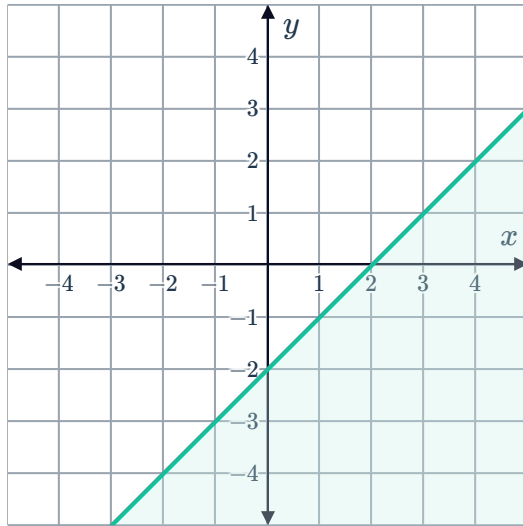
9

a

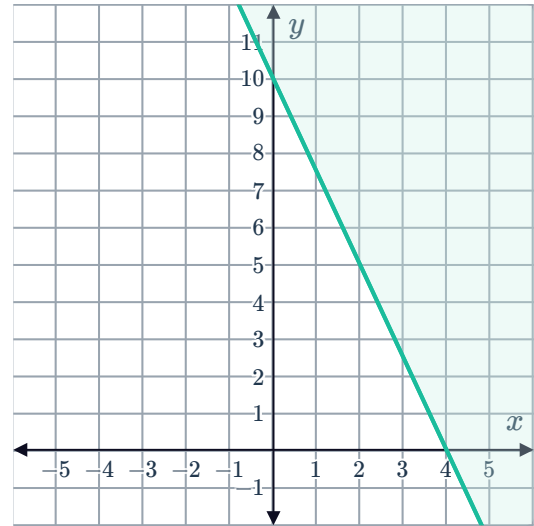
b



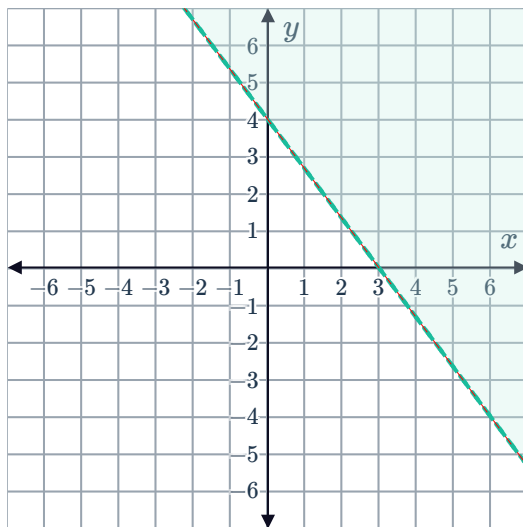
a



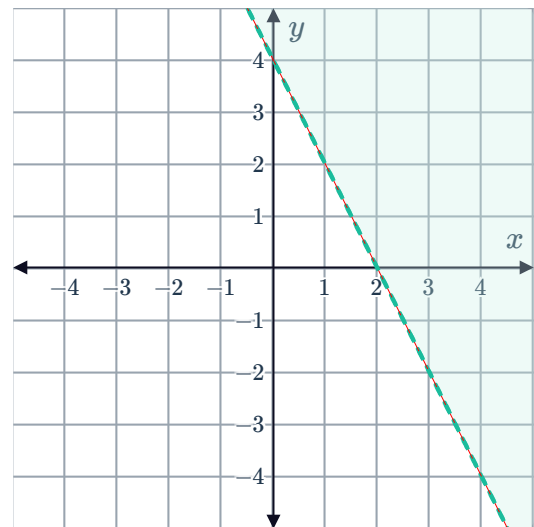
b



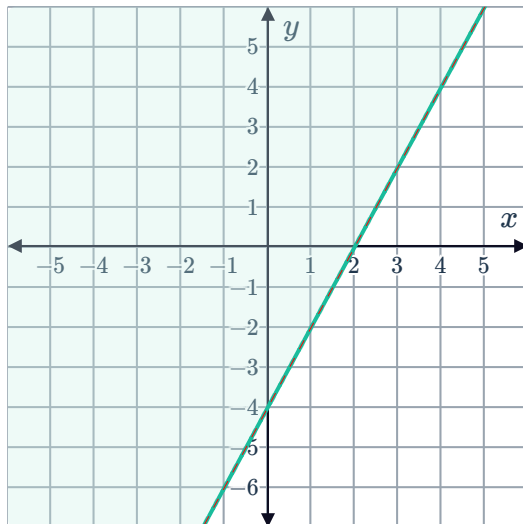
c



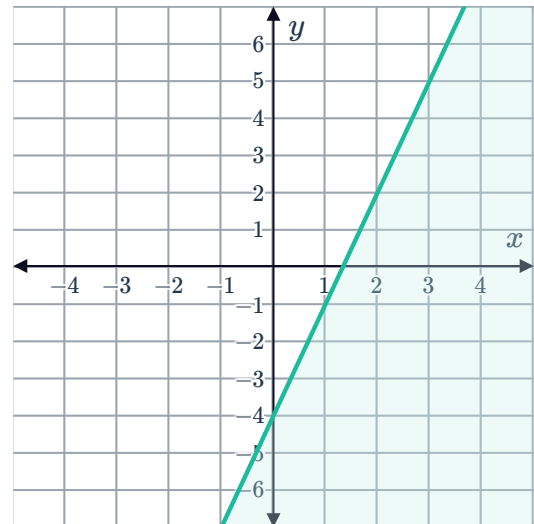
d



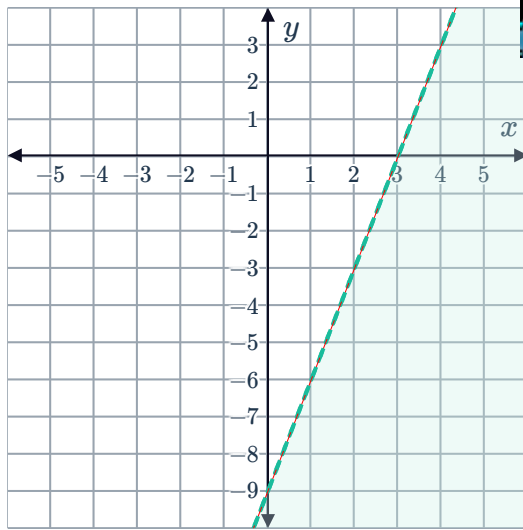
e



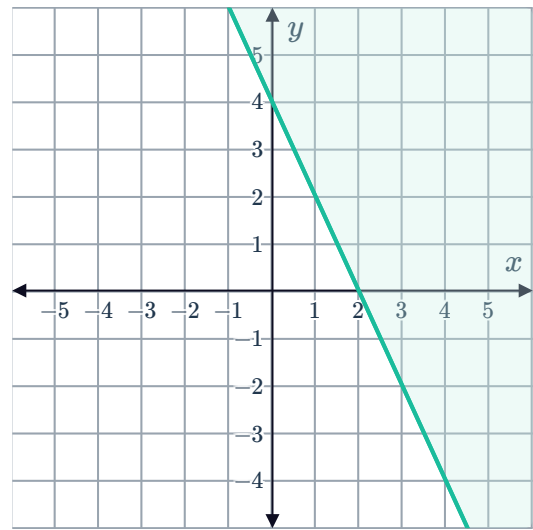
f



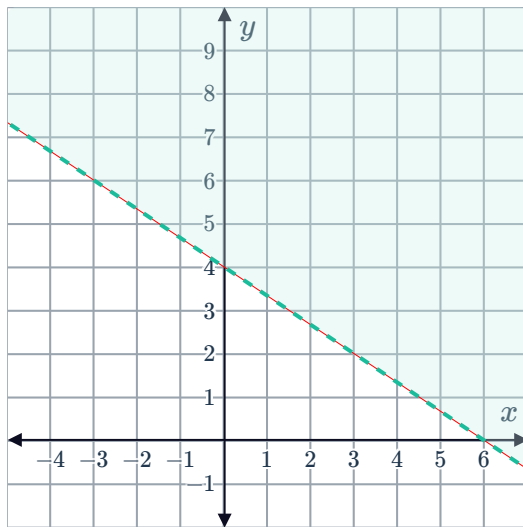
g



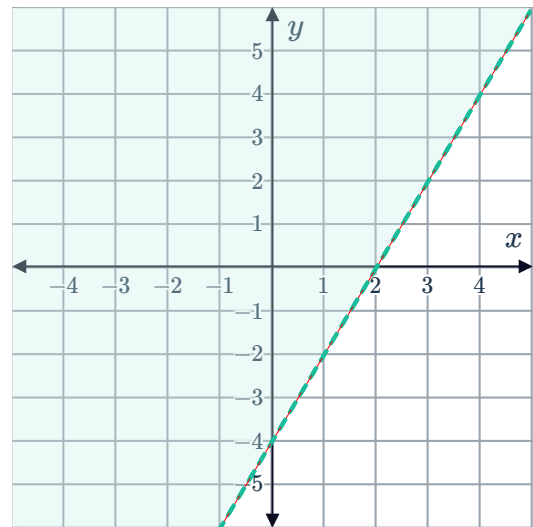
h



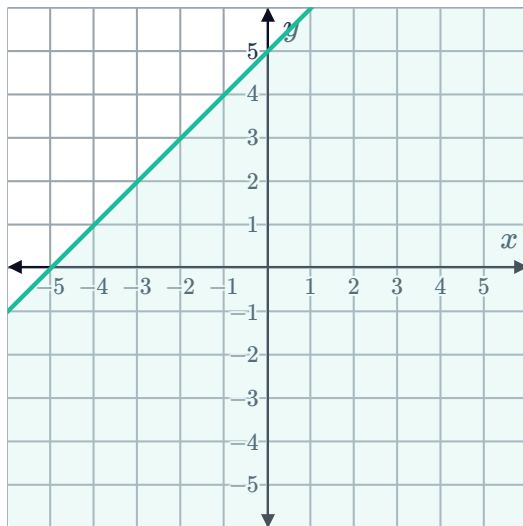
i



j



k



11

a $y \leq 4x - 5$

b $y \geq 3x - 6$

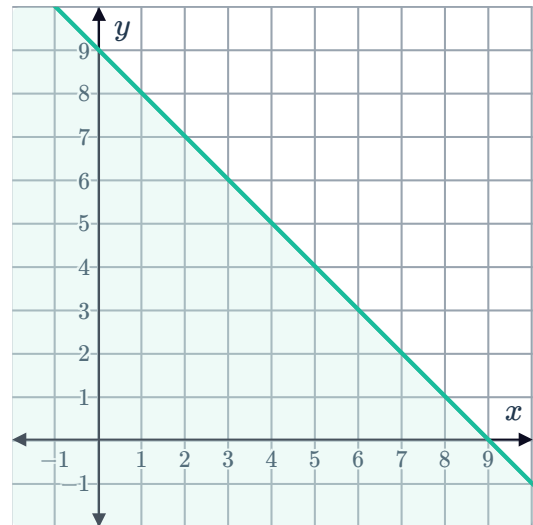


Applications

12

a $y \leq 9 - x$

b



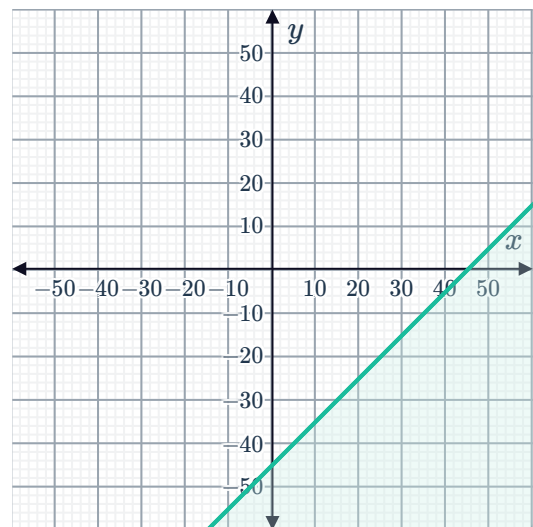
c

- i No ii No iii Yes iv Yes

13

a $y \leq x - 45$

b



c 65 minutes

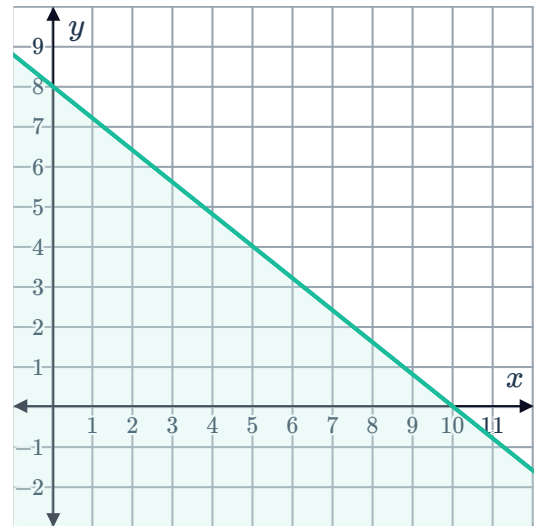
14

a

$$y \leq 8 - \frac{4}{5}x$$



b



c

- i No ii Yes iii No iv Yes